

Problem Set 5.2 – AI Social Media Auditor

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Source Code

```
1  # Raw text needs normalising before counting because Python compares
2  # strings exactly - "AI", "ai", and "AI," are three different keys to a
3  # dict even though they're the same word to a person. Lowercasing and
4  # stripping punctuation collapses all of those into one key.
5  #
6  # set_a - set_b only gives what's unique to A; set_a ^ set_b gives
7  # everything unique to either side combined, without saying which side
8  # it came from. Use the minus version when you need to know which
9  # platform a follower belongs to, and ^ when you just want the total
10 # "not on both" group.
11 #
12 # Author: Kshitiz
13
14 import string
15
16 # --- Task 1: Word Frequency Counter ---
17 text = ("AI is changing the world. ai is powering new tools, and AI is "
18         "helping researchers learn faster. The world loves AI, and the "
19         "world depends on it more each year.")
20
21 word_counts = {}
22 for raw_word in text.split():
23     cleaned_word = raw_word.lower().strip(string.punctuation)
24     if not cleaned_word:
25         continue
26     word_counts.setdefault(cleaned_word, 0)
27     word_counts[cleaned_word] += 1
28
29 print("--- Word Frequency Counter ---")
30 print(word_counts)
31
32 # --- Task 2: Follower Deduplicator ---
33 platform_a_followers = ["u101", "u102", "u103", "u104", "u105"]
34 platform_b_followers = ["u103", "u104", "u106", "u107"]
35
36 set_a = set(platform_a_followers)
37 set_b = set(platform_b_followers)
38
39 both_platforms = set_a & set_b
40 only_a = set_a - set_b
41 only_b = set_b - set_a
42 either_only = set_a ^ set_b
43
44 print()
45 print("--- Follower Deduplicator ---")
46 print(f"Follows on both platforms: {both_platforms}")
47 print(f"Unique to Platform A: {only_a}")
48 print(f"Unique to Platform B: {only_b}")
49 print(f"Unique to exactly one platform: {either_only}")
50
51 # Submission Note: for the sample paragraph, "ai" comes out to 4 (AI, ai,
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```

52 # AI, AI - the trailing comma and period get stripped) and "is"/"the"/
53 # "world" each come to 3, matching a manual word-by-word count of the
54 # text. For the follower lists, u103 and u104 show up on both platforms,
55 # u101/u102/u105 are only on A, and u106/u107 are only on B - the
56 # symmetric difference is just those last two groups combined, which
57 # checks out against adding only_a and only_b together by hand.

```

Trace Log

```

1 --- Word Frequency Counter ---
2 {'ai': 4, 'is': 3, 'changing': 1, 'the': 3, 'world': 3, 'powering': 1, 'new
  ': 1, 'tools': 1, 'and': 2, 'helping': 1, 'researchers': 1, 'learn': 1,
  'faster': 1, 'loves': 1, 'depends': 1, 'on': 1, 'it': 1, 'more': 1, '
  each': 1, 'year': 1}
3
4 --- Follower Deduplicator ---
5 Follows on both platforms: {'u103', 'u104'}
6 Unique to Platform A: {'u101', 'u102', 'u105'}
7 Unique to Platform B: {'u107', 'u106'}
8 Unique to exactly one platform: {'u102', 'u105', 'u101', 'u107', 'u106'}

```

Word counts and set operations match the worked examples in the assignment (set element order differs, which is expected).

Submission Note

For the sample paragraph, "ai" comes out to 4 (AI, ai, AI, AI – the comma and period get stripped) and "is", "the", "world" each come to 3, matching a manual word-by-word count. For the follower lists, u103 and u104 show up on both platforms, u101/u102/u105 are only on A, and u106/u107 are only on B – the symmetric difference is just those last two groups combined, which checks out against adding only_a and only_b together by hand.